

IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

Department of Computer Science & Engineering | School of Engineering & Technology

A SUMMER INTERNSHIP REPORT ON

FRONT-END WEB DEVELOPMENT

AT TRIPEKART (tripekart.com)

Submitted in partial fulfillment of the requirements for the degree of B.Tech in Computer Science & Engineering

SUBMITTED BY:

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Semester / Year: 5th Semester / 3rd Year

Program: B.Tech (CSE / IT)

TRAINING DETAILS:

Company: Tripekart

Website: <http://tripekart.com/>

Duration: May 2026 – August 2026

Role: Front-End Developer Intern

ACADEMIC SESSION: 2026–2027

ACADEMIC APPROVAL CERTIFICATE

This is to certify that the Summer Internship Report entitled "**Front-End Web Development at Tripekart**" is a bonafide record of industrial training carried out by **Akshay Yadav (Roll No: 2410030258)**, a student of 5th Semester B.Tech (CSE) at **IILM University, Greater Noida**.

This training was undertaken at **Tripekart (<http://tripekart.com/>)** during the period from **May 2026 to August 2026** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering.

Faculty Coordinator / Guide
Department of CSE, IILM

Head of Department (CSE)
School of Engineering & Tech

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **IILM University, Greater Noida** for providing me the platform and support to undergo industrial training as part of my B.Tech curriculum.

I am deeply thankful to the engineering team at **Tripekart (<http://tripekart.com/>)** for providing me the opportunity to work as a Front-End Web Developer Intern from May 2026 to August 2026. Working on production-level travel products provided immense learning exposure.

I also extend my thanks to my faculty guide and the Head of Department, Computer Science & Engineering, IILM University, for their continuous support and suggestions during the preparation of this report.

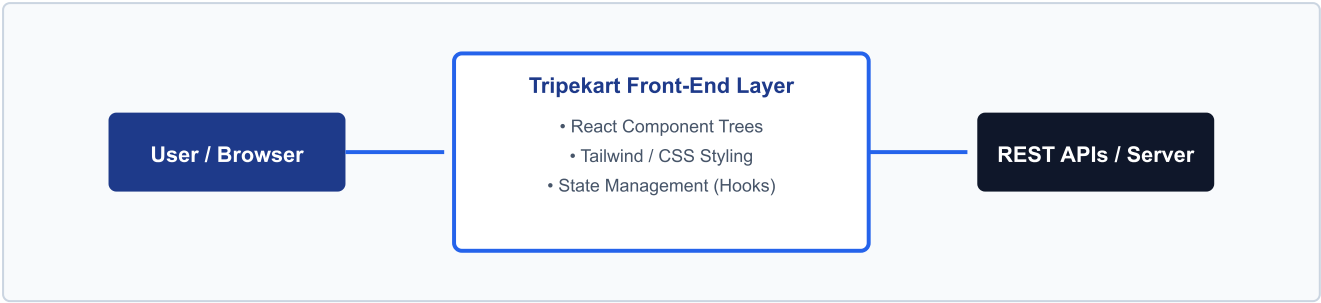
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1. COMPANY PROFILE — TRIPEKART

Tripekart (<http://tripekart.com/>) is an online travel and tourism platform specializing in customized itineraries, group backpacking trips, weekend getaways, and adventure tours across destinations like Ladakh, Spiti Valley, Meghalaya, and Himachal Pradesh. The portal offers users transparent pricing, day-wise trip schedules, dynamic package filtering, and easy booking inquiries.

Organization Domain: Travel & Tourism Tech | E-Commerce Booking Engine
Technical Focus: Responsive UI/UX, Dynamic Itinerary Rendering, High Performance Media Handling, Lead Capture Systems.

Tripekart Web System Architecture Overview:



2. INTRODUCTION & OBJECTIVES

As a 3rd-year B.Tech CSE student at IILM University, my 4-month summer internship at Tripekart focused on front-end web development. In e-commerce travel applications, page loading speed, dynamic responsiveness, and clear visual hierarchy directly drive user engagement and booking conversions.

- **Responsive Interface Design:** Build user-friendly UI components for trip packages, detail pages, and inquiry forms.
- **Cross-Device Compatibility:** Ensure seamless navigation across mobile, tablet, and desktop screens using modern CSS layouts.
- **API Integration:** Connect dynamic client-side views with backend RESTful endpoints for real-time pricing and booking workflows.
- **Code Efficiency:** Minimize re-renders, optimize visual media, and write clean, modular React component code.

3. TOOLS & TECH STACK USED

During the internship, I utilized the following modern web technologies and software tools:

Technology / Tool	Category	Key Application at Tripekart
HTML5 & CSS3	Core Web Tech	Semantic structural markup, grid/flexbox layouts, responsive media queries.
JavaScript (ES6+)	Programming Language	Asynchronous data fetch calls, DOM manipulation, dynamic form validations.
React.js	Front-End Framework	Building reusable UI components, routing, and component state management.
Tailwind CSS	Styling Framework	Rapid utility-first responsive styling and consistent UI theme design.
Git & GitHub	Version Control	Feature branching, tracking pull requests, and resolving code merges.
Postman	API Testing Tool	Testing REST API payloads and verifying backend endpoints before integration.
Chrome DevTools	Debugging Suite	DOM inspection, performance profiling, mobile viewport testing, and network analysis.

4. MONTH-WISE WORK SUMMARY (MAY 2026 – AUGUST 2026)

MAY 2026: Onboarding & Component Setup - Analyzed existing codebase of Tripekart. - Developed basic UI elements: customized button sets, navbar, and sticky footers. - Standardized mobile-first breakpoint rules across package landing pages.	JUNE 2026: Interactive Modules & Forms - Redesigned trip inquiry and lead capture forms. - Implemented regex-based client-side validation for phone numbers and email inputs. - Developed dynamic accordion dropdowns for day-wise itinerary views.
JULY 2026: API Integration & Filtering - Connected package listing views to live REST API endpoints. - Implemented custom search and dynamic filter controls (filter by budget, duration). - Managed loading state indicators and modal booking popups.	AUGUST 2026: Performance & Testing - Optimized image assets by implementing native lazy loading (`loading="lazy"`). - Conducted cross-browser compatibility tests on Chrome, Edge, and Safari. - Fixed responsive alignment bugs and prepared developer documentation.

5. TECHNICAL IMPLEMENTATION & MODULES DEVELOPED

One of the core modules I worked on at Tripekart was the **Interactive Itinerary Accordion Component** for tour package details pages (e.g., Spiti Valley Backpacking). This component allows users to expand/collapse day-by-day schedules cleanly without triggering full page re-renders.

Sample Component Code Snippet (React.js + Tailwind CSS):

```
import React, { useState } from 'react'; // Reusable Day-wise Itinerary Accordion
Component const ItineraryAccordion = ({ dayNumber, title, description, highlights }) => {
  const [isOpen, setIsOpen] = useState(false); return ( <div className="border-b border-slate-
  200 my-2 rounded overflow-hidden"> <button onClick={() => setIsOpen(!isOpen)} className="w-
  full text-left py-3 px-4 bg-slate-50 hover:bg-slate-100 flex justify-between items-center
  transition-colors"> <span className="font-semibold text-slate-800"> Day {dayNumber}: {title}
  </span> <span className="text-blue-600 font-bold text-lg"> {isOpen ? '-' : '+'} </span>
  </button> {isOpen && ( <div className="p-4 bg-white text-slate-600 text-sm"> <p
  className="mb-2">{description}</p> {highlights && ( <div className="text-xs bg-blue-50 p-2
  rounded text-blue-800 font-medium"> Key Highlights: {highlights} </div> )} </div> )} </div>
  ); }; export default ItineraryAccordion;
```

Technical Highlight: Using state-driven conditional rendering reduced unnecessary DOM node allocation, improving scroll responsiveness on low-end mobile devices during travel itinerary browsing.

6. KEY CHALLENGES & SOLUTIONS

1. Image Heavy Pages & Performance Bottlenecks

Challenge: Travel websites rely heavily on high-res gallery images. Initial page loads were slow on mobile network connections.

Solution: Implemented native image lazy-loading (`loading="lazy"`), converted raster assets to WebP format, and fixed aspect ratios to eliminate cumulative layout shifts (CLS).

2. Dynamic Form State & Validation Handling

Challenge: Multi-step lead capture forms allowed invalid contact inputs, resulting in degraded lead quality.

Solution: Written client-side regular expression (Regex) checks for mobile numbers and email addresses, disabling submission buttons until all required constraints passed.

7. LEARNINGS & CONCLUSION

My 4-month summer internship at **Tripekart** provided practical exposure to production-grade front-end development. Key technical and professional takeaways include:

- **Production Code Standard:** Gained hands-on experience structuring component trees, keeping UI logic modular, and maintaining clean folder structures.
- **UI/UX Empathy:** Understood how minor front-end enhancements—such as fast button feedback and clear error hints—directly impact user engagement.
- **Industry Workflow:** Learned real-world team practices including Git branch management, daily standups, sprint tracking, and code reviews.

In conclusion, this industrial training fulfilled the B.Tech CSE learning goals at IILM University, bridging classroom fundamentals with industry execution.

8. REFERENCES

1. Tripekart Official Web Portal — <http://tripekart.com/>
2. MDN Web Docs (JavaScript, HTML5 & CSS Layouts) — <https://developer.mozilla.org>
3. React.js Official Documentation — <https://react.dev>
4. Tailwind CSS Documentation — <https://tailwindcss.com>